

# Morning quantitative update

September 11, 2026 | saved results checked after the 09:36 EDT restart

**Useful repairs and some large fit improvements, but no finished calibration or certified policy package.**

## What is established

The approved parenthood-only housing floor and equivalence scale are implemented in the isolated sequential model. The initial steady state reproduces exactly and balances Social Security at the fixed 17.9% payroll tax. A separate terminal steady-state test also balances; its preference change remains diagnostic.

The 19-case sensitivity panel finished successfully. The 24-proposal joint round produced 23 valid candidates, including the twice-reproduced smoke point; one candidate failed the unchanged strict housing-equilibrium check. No failed candidate was admitted.

## The economic result that matters

The tested joint candidate raises the first-birth room response from 0.439 to 0.692, against 0.720 in the data. Mean rooms improve from 6.246 to 5.609, against 5.561; ownership at ages 30-55 rises from 57.48% to 63.10%, against 64.83%.

However, its recent-parent ownership gap worsens from -5.32 to -6.34 percentage points, against +16.29 in the data. This row is retained. Once all 12 moments are counted, loss falls only 4.8%: 1499.85 to 1428.17. Recent-parent ownership accounts for 84.3% of baseline loss and 97.0% of the joint candidate loss. This candidate is not a selected final estimate.

## Transition and interruption

The last collected six-date transition iterate reduced maximum housing imbalance from 45.3% to 1.76%, still above the 0.02% gate. Pension residuals were below  $4.4e-10$ . The final result has not been collected: cluster authentication was unavailable at 09:36. This does not establish a completed perfect-foresight path, an estimated historical shock, or policy effects.

Agent work stopped when credits ran out around 03:27. Previously submitted cluster jobs could finish independently; repeated automatic wake-ups are not evidence of further research. Work has resumed with one lead and hourly monitoring. Next: collect the transition, finish recent-parent measurement for all joint candidates, then continue the full-objective search after its exact-loop check.

# Starting point: complete fit

All 13 restrictions: 12 scored moments and a separate normalization. Proportions and ownership gaps are in fraction units; 0.01 equals one percentage point.

| Moment                                | Target   | Model      | Gap         | Weight      | Loss      |
|---------------------------------------|----------|------------|-------------|-------------|-----------|
| Initial fertility normalization       | 2.1      | 2.10003    | 3.28629e-05 | -           | -         |
| Childlessness, women 40-44            | 0.198279 | 0.180585   | -0.0176937  | 35532.3     | 11.124    |
| Exactly one, among mothers 40-44      | 0.213655 | 0.233236   | 0.0195807   | 26952.8     | 10.3339   |
| Mean age at first birth               | 25.9763  | 26.5851    | 0.608877    | 139.828     | 51.8386   |
| First births at ages 30+              | 0.249278 | 0.265168   | 0.0158896   | 13866.1     | 3.50088   |
| Wealth / annual gross earnings        | 6.14586  | 6.09688    | -0.0489844  | 7.5951      | 0.0182242 |
| Annual bequests / wealth              | 0.0088   | 0.012368   | 0.00356803  | 5.16529e+06 | 65.7586   |
| Old wealth-income ratio: p90 / median | 3.51594  | 3.72976    | 0.213821    | 10.6164     | 0.485372  |
| Mean occupied rooms, capped at 9      | 5.5611   | 6.24638    | 0.685285    | 128.021     | 60.1205   |
| Ownership, household heads 30-55      | 0.648334 | 0.574768   | -0.0735657  | 2339.36     | 12.6604   |
| First-birth room response             | 0.720246 | 0.438847   | -0.2814     | 137.565     | 10.8932   |
| Rooms: 3+ versus 1-2 dependents       | 0.347067 | 0.166215   | -0.180852   | 280.528     | 9.17532   |
| Recent-parent ownership gap           | 0.162896 | -0.0532436 | -0.216139   | 27055.8     | 1263.94   |

Total working minimum-distance loss: **1499.85**. The fertility normalization has no weight or loss contribution.

Weights are fixed working choices, not an efficient GMM covariance matrix. CPS uses approximate pooled scales; NCHS uses annual 2003-2006 variation as a typical-year discrepancy scale. Housing and wealth use retained bootstrap scales; bequests retain a synthetic 5% scale.

Recent-parent ownership uses actual current births into previously empty-dependent homes, compared with current empty homes including former parents. The synchronized snapshot and dependent-residence approximation are maintained explicitly; this is not an exact annual ACS interview reconstruction.

# Tested joint candidate: complete fit

All 13 restrictions: 12 scored moments and a separate normalization. Proportions and ownership gaps are in fraction units; 0.01 equals one percentage point.

| Moment                                | Target   | Model      | Gap         | Weight      | Loss        |
|---------------------------------------|----------|------------|-------------|-------------|-------------|
| Initial fertility normalization       | 2.1      | 2.10028    | 0.000276518 | -           | -           |
| Childlessness, women 40-44            | 0.198279 | 0.205085   | 0.00680612  | 35532.3     | 1.64597     |
| Exactly one, among mothers 40-44      | 0.213655 | 0.202669   | -0.0109862  | 26952.8     | 3.25312     |
| Mean age at first birth               | 25.9763  | 25.9787    | 0.0024364   | 139.828     | 0.000830027 |
| First births at ages 30+              | 0.249278 | 0.227356   | -0.0219225  | 13866.1     | 6.66396     |
| Wealth / annual gross earnings        | 6.14586  | 5.8135     | -0.332365   | 7.5951      | 0.839005    |
| Annual bequests / wealth              | 0.0088   | 0.0106133  | 0.00181335  | 5.16529e+06 | 16.9847     |
| Old wealth-income ratio: p90 / median | 3.51594  | 4.07964    | 0.563702    | 10.6164     | 3.37345     |
| Mean occupied rooms, capped at 9      | 5.5611   | 5.60933    | 0.0482294   | 128.021     | 0.297786    |
| Ownership, household heads 30-55      | 0.648334 | 0.631009   | -0.0173249  | 2339.36     | 0.702167    |
| First-birth room response             | 0.720246 | 0.691588   | -0.0286583  | 137.565     | 0.112982    |
| Rooms: 3+ versus 1-2 dependents       | 0.347067 | 0.173255   | -0.173812   | 280.528     | 8.47488     |
| Recent-parent ownership gap           | 0.162896 | -0.0634248 | -0.22632    | 27055.8     | 1385.82     |

Total working minimum-distance loss: **1428.17**. The fertility normalization has no weight or loss contribution.

Weights are fixed working choices, not an efficient GMM covariance matrix. CPS uses approximate pooled scales; NCHS uses annual 2003-2006 variation as a typical-year discrepancy scale. Housing and wealth use retained bootstrap scales; bequests retain a synthetic 5% scale.

Recent-parent ownership uses actual current births into previously empty-dependent homes, compared with current empty homes including former parents. The synchronized snapshot and dependent-residence approximation are maintained explicitly; this is not an exact annual ACS interview reconstruction.

# Parameters and remaining claims

All 17 parameter/restriction rows. Nine structural coordinates vary jointly. The initial child-utility intercept is separately normalized to 2.1; fixed and budget-derived objects are not additional searched coordinates.

| Parameter                  | Starting | Joint    | Bounds / restriction | Near bound? |
|----------------------------|----------|----------|----------------------|-------------|
| Annual discount factor     | 0.995277 | 0.995157 | [0.94, 0.9995]       | No          |
| First-birth taste scale    | 2.16817  | 1.48095  | [0.02, 50]           | No          |
| Later-birth taste scale    | 1.77077  | 1.47462  | [0.02, 50]           | No          |
| Tenure taste parameter     | 1.05373  | 1.09752  | [0.1, 5]             | No          |
| Housing-supply scale       | 8.43484  | 6.61785  | [0.2, 80]            | No          |
| Bequest level              | 0.57035  | 0.345935 | [0, 8]               | No          |
| Bequest curvature          | 0.103724 | 0.137972 | [0.02, 16]           | Yes         |
| First-birth fixed cost     | 4.61973  | 4.20301  | [0, 8]               | No          |
| Parenthood housing floor   | 0.71685  | 1.18189  | [0.1, 2.3]           | No          |
| Additional per-child floor | 0        | 0        | Fixed                | -           |
| Child-utility intercept    | 0.282704 | 0.309861 | Normalized           | -           |
| Payroll tax                | 0.179    | 0.179    | Fixed                | -           |
| Period pension             | 2.04636  | 2.04636  | Budget derived       | -           |
| Supply elasticity          | 0.63     | 0.63     | Fixed                | -           |
| Tenure shock scale         | 0.005    | 0.005    | Fixed                | -           |
| Consumption share          | 0.733    | 0.733    | Fixed                | -           |
| CRRA curvature             | 2        | 2        | Fixed                | -           |

Annual beta enters a four-year period as beta to the fourth power. Near-bound flags use the inherited rule: within 1% of the stated bound span. The local 12-by-9 Jacobian has numerical rank 9 but condition number about 1856; this is not evidence of strong identification.

Source boundary: initial\_sensitivity\_panel and initial\_joint\_round\_01/smoke\_readout preserve the original solved packets. initial\_calibration\_contract/saved\_case\_scores contains complete machine-readable fits, parameters, source hashes and scoring receipts. Original model/observer diagnostic flags remain unchanged; these working scores do not certify a production benchmark.